MACHINE LEARNIN

1. What is the most appropriate no. of clusters for the data points represented by the following dendrogram:

c) 4

2. In which of the following cases will K-Means clustering fail to give good results?

d) 1, 2 and 4

3. The most important part of is selecting the variables on which clustering is based.

d) formulating the clustering problem

4. The most commonly used measure of similarity is the or its square.

a) Euclidean distance

5. -----is a clustering procedure where all objects start out in one giant cluster. Clusters are formed by dividing this cluster into smaller and smaller clusters.

b) Divisive clustering

6. Which of the following is required by K-means clustering?

d) All answers are correct

7. The goal of clustering is to

d) All of the above

8. Clustering is a

b) Unsupervised learning

9. Which of the following clustering algorithms suffers from the problem of convergence at local optima?

a) K- Means clustering

10. Which version of the clustering algorithm is most sensitive to outliers?

a) K-means clustering algorithm

11. Which of the following is a bad characteristic of a dataset for clustering analysis

d) All of the above

12. For clustering, we do not require

a) Labeled data

13. How is cluster analysis calculated?

• Partitioning algorithms: Construct various partitions and then evaluate them by some criterion

• Hierarchy algorithms: Create a hierarchical decomposition of the set of data (or objects) using some criterion

• Density-based: based on connectivity and density functions

• Grid-based: based on a multiple-level granularity structure

• Model-based: A model is hypothesized for each of the clusters and the idea is to find the best fit of that model to each other

14. How is cluster quality measured?

To measure the quality of a clustering, we can use the average silhouette coefficient value of all objects in the data set.

15. What is cluster analysis and its types?

The clustering analysis can be broadly divided into two categories: Agglomerative and Divisive.

Agglomerative: This approach first considers all the points as individual clusters and then finds out the similarity between two points, puts them into a cluster. Then it goes on finding similar points and clusters until there is only one cluster left i.e., all points belong to a big cluster. This is also called the bottom-up approach.

Divisive: It is opposite of the agglomerative approach. It first considers all the points to be part of one big cluster and in the subsequent steps tries to find out the points/ clusters which are least similar to each other and then breaks the bigger cluster into smaller ones. This continues until there are as many clusters as there are datapoints. This is also called the top-down approach.